

CSS Layout, Tailwind CSS & Advanced Styling

Block & Inline • div/span • Tailwind Utility-First • Flex • Grid • Effects

What We'll Cover Today

01

Layout Basics

div, span, id/class, block vs inline, comments

02

Tailwind CSS Intro

Utility-first, CDN setup, responsive, dark mode

03

Display & Positioning

display, z-index, box-sizing, float, overflow

04

Flexbox

flex, flex-col, justify, items, gap, wrap

05

CSS Grid

grid-cols, col-span, place-items

06

Spacing & Sizing

m-, p-, space-x/y-, w-, h-

07

Typography & Backgrounds

font-, text-, bg-, border-, rounded-

08

Effects & Extras

shadow, opacity, blur, grayscale, ring, divide

01

Layout Basics

div, span, id/class attributes, block vs inline, HTML comments

div vs span — The Two Containers

<div> Block Container

- Starts on a new line
- Takes full available width
- Groups sections for layout
- Used with CSS for page structure

 Inline Container

- Stays within text flow
- Only takes content-wide space
- Styles part of a sentence
- Cannot contain block elements

index.html

```
<div style="background: #e0f7fa; padding: 12px;">
  This is a block div — full width.
  <span style="color: red; font-weight: bold;">red bold</span>
  continues inline here.
</div>
<div style="background: #fff9c4; padding: 12px;">
  Second div → starts on a NEW LINE.
</div>
```

● ● ● Output

This is a block div — full width.

red bold continues inline here.

Second div → starts on a NEW LINE.

	Block	Inline
New line?	✓ Yes	✗ No
Width	Full width	Content only

Common HTML Attributes — id, class, style, src, href, alt

id

Unique identifier

```
<h1 id="main">Title</h1>
```

Used in CSS as #main, in JS via getElementById

class

Reusable grouping

```
<p class="highlight">Text</p>
```

Used in CSS as .highlight — multiple elements

style

Inline CSS styling

```
<p style="color:blue;">Blue</p>
```

Quick styling; prefer external CSS for scale

src

Source file path / URL

```

```

Used in img, script, iframe, video tags

href

Hyperlink destination URL

```
<a href="https://site.com">Go</a>
```

Used in anchor <a> and link <link> tags

alt

Alt text for images

```

```

Accessibility + SEO — always include this!

HTML Comments & Block vs Inline Elements — Full Reference

HTML Comments <!-- -->

Not rendered on page • Used for notes, debug, and TODO reminders

```
<!-- This is a comment — invisible to users -->
<h1>Title</h1>  <!-- Main heading -->
<!-- TODO: add form here -->
```

Block Elements

Start on new line • Take full width
Used for structural layout

```
<div>
<p>
<h1> to <h6>
<section>
<article>
<header>
<footer>
<ul> <ol> <li>
```

Inline Elements

Stay within text flow • Content-wide only
Used for styling within block elements

```
<span>
<a>
<strong>  <em>
<img>
<label>
<input>
<button>
```



Tip: Block elements can contain inline elements, but inline elements should NOT wrap block elements.

02

Tailwind CSS

Utility-first framework • Setup • Responsive • Dark Mode

What is Tailwind CSS? — Utility-First Explained

Traditional CSS approach

```
.card {  
  padding: 1rem;  
  background-color: white;  
  border-radius: 0.5rem;  
  box-shadow: 0 2px 8px gray;  
}
```

→ Write custom CSS in a separate file

Tailwind Utility-First approach

```
<div class="p-4 bg-white rounded-lg shadow-md">  
  <!-- content -->  
</div>
```

- Apply pre-built classes directly in HTML
- No context switching, no naming conflicts



Single-purpose classes

Each class does ONE thing: p-4 = padding 1rem, text-red-500 = red text



No naming required

Never think of a class name like .card-wrapper-inner-left again



Consistent design system

Spacing, colors, sizes follow a fixed scale — automatic consistency



Responsive built-in

Add sm: md: lg: prefixes to apply styles at specific breakpoints

Tailwind Setup — CDN (Quick Start)

Quickest way to use Tailwind: add one `<script>` tag in your HTML file's `<head>`. No npm, no build step needed.

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1.0">
  <title>My Tailwind Page</title>

  <!-- Add this ONE line to use Tailwind via CDN -->
  <script src="https://cdn.tailwindcss.com"></script>
</head>
<body class="bg-gray-100 p-8">

  <h1 class="text-3xl font-bold text-teal-600">
    Hello, Tailwind!
  </h1>

  <p class="mt-4 text-gray-600 text-lg">
    This paragraph uses utility classes.
  </p>
```

Class breakdown

<code>bg-gray-100</code>	→ light gray background
<code>p-8</code>	→ padding 2rem all sides
<code>text-3xl</code>	→ font-size 1.875rem
<code>font-bold</code>	→ font-weight: 700
<code>text-teal-600</code>	→ teal text color
<code>mt-4</code>	→ margin-top 1rem
<code>px-6 py-2</code>	→ horiz/vert padding
<code>rounded-lg</code>	→ border-radius 0.5rem
<code>hover:bg-teal-600</code>	→ darker on hover

Preview
Hello, Tailwind!

`rounded-lg hover:bg-teal-600">`

Responsive Utilities — sm: md: lg: xl:

Prefixes apply styles at specific screen widths — mobile-first by default (no prefix = all sizes).

(default)	0px+	All screens (mobile-first base)
sm:	640px+	Small screens and up
md:	768px+	Medium screens (tablets) and up
lg:	1024px+	Large screens (laptops) and up
xl:	1280px+	Extra large screens and up

responsive.html

```
<!-- Text size changes per breakpoint -->
<p class="text-sm md:text-lg lg:text-2xl">
  I grow at md and lg screens!
</p>
```

```
<!-- Columns: 1 on mobile, 2 on tablet, 3 on desktop -->
```

• • • Responsive behavior

Mobile (< 640px):
text-sm, grid-cols-1, nav hidden

Tablet (≥ 768px):
text-lg, grid-cols-2, nav hidden

Dark Mode Support — Tailwind dark: prefix

Enable darkMode: 'class' in config → then use dark: prefix on any utility class. Toggle by adding/removing 'dark' class on <html>.

```
darkmode.html
<!DOCTYPE html>
<html lang="en" class="light">
<head>
  <script src="https://cdn.tailwindcss.com"></script>
  <script>
    tailwind.config = { darkMode: 'class' };

    function toggleDarkMode() {
      document.documentElement.classList.toggle('dark');
    }
  </script>
</head>
<body class="bg-white text-black dark:bg-gray-900 dark:text-white
            min-h-screen flex flex-col items-center justify-
center">

  <h1 class="text-3xl font-bold mb-4">
    Hello, Dark Mode!
  </h1>

  <button
    onclick="toggleDarkMode()"
    class="px-4 py-2 bg-blue-500 text-white rounded hover:bg-
    blue-600">
    Toggle Dark Mode
  </button>
```



Light Mode

Hello, Dark Mode!

[Toggle Dark Mode]



Dark Mode

Hello, Dark Mode!

[Toggle Dark Mode]

class= light → dark:

bg-white → dark:bg-gray-900

03

Display, Box-Sizing, z-index Overflow, Float & Clear

Core layout control properties

Display Property & Box-Sizing

display — Controls how element is shown

block	Full width, new line	<code><div class="block"></code>
inline	Inline, content width	<code></code>
flex	Flex container	<code><div class="flex"></code>
grid	Grid container	<code><div class="grid"></code>
hidden	Hides the element	<code><div class="hidden"></code>

tailwind.html

```
<!-- Display examples -->
<div class="flex">Flex Row</div>
<div class="hidden">Not visible</div>
<div class="block">Block, new line</div>
```

box-sizing — How width is calculated

content-box (default):

width = content only. Padding & border ADD to total size.

```
<div class="box-content w-40 p-4 border">
```

border-box (recommended):

width includes content + padding + border. Box STAYS the same size.

```
<div class="box-border w-40 p-4 border">
```

content-box
w=160, total=192px

border-box
w=160, total=160px

← same w="160", different total size →

z-index, Overflow, Float & Clear

z-index — Stacking Order

Controls which element appears ON TOP when elements overlap.
Higher z-index = in front. Only works with position set.

```
<div class="relative">
  <h1 class="z-50 absolute flex
    justify-center items-center">
    On top ← z-50
  </h1>
  
</div>
```

float — Element Alignment

Moves element left/right; text wraps around it.

```

<p>Text wraps to the right of the image...</p>

<p>This text wraps left of the image...</p>
```

overflow — Content Clipping

<code>overflow-hidden</code>	Clips content, nothing shows outside
<code>overflow-scroll</code>	Always shows scrollbars
<code>overflow-auto</code>	Scrollbars only when needed
<code>overflow-visible</code>	Content spills outside (default)

clear — Stop Float Wrap

Prevents element from wrapping next to floated elements.

```
<!-- Without clear: content overlaps floats -->

<p>Text wraps here...</p>

<!-- clear-both: stops the wrap -->
<div class="clear-both"></div>
<p>This starts below all floats cleanly.</p>
```

`clear-left` / `clear-right` / `clear-both` → stops wrapping from left, right, or both sides.

CSS Positioning — static, relative, absolute, fixed, sticky

static

Default — follows normal document flow. Cannot use top/left/z-index.

relative

Offset from its OWN normal position. Other elements don't move.

absolute

Positioned relative to nearest positioned ancestor (not static).

fixed

Fixed to the viewport — doesn't scroll with the page.

sticky

Sticks to its scroll position within its parent. Toggles between relative/fixed.

positioning.html

```
<div class="h-[900px]">
  <div class="static bg-yellow-200">I'm static</div>
  <div class="relative top-14 left-4 bg-green-200">
    I'm relative (moved 3.5rem down, 1rem right)
  </div>
</div>
```

Page layout

static

relative

absolute inside

sticky

04

Flexbox

flex flex-col justify-* items-* gap-* flex-wrap

Flexbox Fundamentals — flex, flex-col, gap

flex → Horizontal Row (default)

```
<div class="flex">
  <div>Box 1</div>
  <div>Box 2</div>
  <div>Box 3</div>
</div>
```

Box 1

Box 2

Box 3

→ items
side by
side

flex-col → Vertical Stack

```
<div class="flex flex-col">
  <div>Box A</div>
  <div>Box B</div>
</div>
```

Box A

Box B

→ stacked vertically

justify-start

Items packed to start (left in row)

justify-center

Items centered on main axis

justify-between

Space between items, none at edges

items-start

Align to top (cross axis)

items-center

Center on cross axis

items-end

Align to bottom

gap-4

1rem gap all sides

gap-x-4

Horizontal gap only

gap-y-4

Vertical gap only

Flexbox — flex-wrap & Complete Example

flex-wrap / flex-nowrap

flex-wrap: items wrap to next line if container is too small

flex-nowrap: items stay on one line (may overflow)

```
<div class="flex flex-wrap gap-2">
  <div class="w-40 bg-pink-300">Box</div>
  <div class="w-40 bg-yellow-300">Box</div>
  <div class="w-40 bg-green-300">Box</div>
</div>
```

Wrapping visual

Box 1

Box 2

Box 3

Box 4

navbar.html

```
<!-- Complete responsive navbar example -->
<nav class="flex flex-wrap justify-between items-center
  gap-4 p-4 bg-white shadow">

  <div class="font-bold text-xl text-teal-600">MyBrand</div>

  <ul class="flex gap-6 text-gray-600">
    <li class="hover:text-teal-500 cursor-pointer">Home</li>
    <li class="hover:text-teal-500 cursor-pointer">About</li>
    <li class="hover:text-teal-500 cursor-pointer">Contact</li>
  </ul>

  <button class="px-4 py-2 bg-teal-500 text-white
    rounded-lg hover:bg-teal-600">
```

Browser Output

MyBrand

Home

About

Contact

Sign Up

- ← flex justify-between keeps items at opposite ends
- ← items-center aligns them vertically centered
- ← gap-4 adds consistent spacing

05

CSS Grid

grid-cols-* col-span-* place-items-*

CSS Grid — grid-cols, col-span, place-items

grid-cols-*

Defines number of equal columns

grid grid-cols-3 gap-4 → 3-column grid

col-span-*

Item spans multiple columns

col-span-2 → takes up 2 column widths

place-items-*

Align items both axes at once

place-items-center → center everything

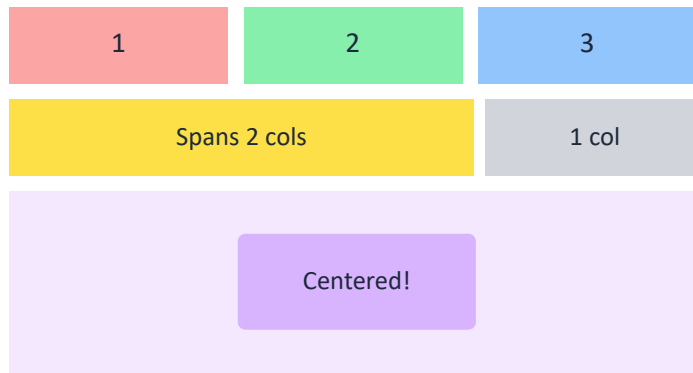
grid.html

```
<!-- 3-column grid -->
<div class="grid grid-cols-3 gap-4">
  <div class="bg-red-200">1</div>
  <div class="bg-green-200">2</div>
  <div class="bg-blue-200">3</div>
</div>

<!-- col-span: spanning multiple columns -->
<div class="grid grid-cols-3 gap-2 pt-10">
  <div class="col-span-2 bg-yellow-300">Spans 2 cols</div>
  <div class="bg-gray-300">1 col</div>
</div>
```

```
<!-- place-items-center: center everything -->
```

● ● ● Output



06

Spacing & Sizing

m-* p-* space-x-* space-y-* w-* h-*

Spacing — m-*, p-*, space-x-*, space-y-*

m-*

Margin (outside)

m-4=1rem all sides mx-4=horizontal my-4=vertical mt-4=top

p-*

Padding (inside)

p-4=1rem all sides px-6=horiz py-2=vert pt-8=top

space-x-*

Horiz gap between children

<div class="flex space-x-4"> adds gap between flex children (horizontal)

space-y-*

Vert gap between children

<div class="flex flex-col space-y-4"> adds gap between stacked children

spacing.html

```
<div class="m-8 p-6 bg-sky-100">  
  Outer margin 2rem, inner padding 1.5rem
```

● ● ● Visual spacing



Width & Height — w-*, h-*

w-full

width: 100% of parent

w-screen

width: 100vw (viewport)

w-1/2

width: 50% of parent

w-1/3

width: 33.333%

w-48

width: 12rem (192px)

w-64

width: 16rem (256px)

h-full

height: 100% of parent

h-screen

height: 100vh (viewport)

h-24

height: 6rem (96px)

h-48

height: 12rem (192px)

min-w-0

min-width: 0

max-w-xl

max-width: 36rem

sizing.html

```
<!-- Full width container, fixed height -->
<div class="w-full h-24 bg-teal-100">Full width, h=6rem</div>
```

```
<!-- Half width, auto height -->
<div class="w-1/2 bg-purple-200 p-4">I'm 50% wide</div>
```

```
<!-- Viewport-filling hero section -->
<section class="w-screen h-screen bg-gray-900 text-white
  flex items-center justify-center">
```

```
  Hero Section
</section>
```

● ● ● Sizing preview

w-full (100%)

w-1/2

w-48

07

Typography, Backgrounds & Borders

font-* text-* bg-* border-* rounded-* ring-* divide-*

Typography & Text Utilities — font-*, text-*

font-bold

font-weight: 700

```
<p class="font-bold">Bold</p>
```

text-center

text-align: center

```
<p class="text-center">Centered</p>
```

font-light

font-weight: 300

```
<p class="font-light italic">Light Italic</p>
```

uppercase

text-transform: uppercase

```
<p class="uppercase">hello → HELLO</p>
```

text-2xl

font-size: 1.5rem

```
<p class="text-2xl">Large text</p>
```

tracking-wide

letter-spacing: 0.025em

```
<p class="tracking-wide">Spaced letters</p>
```

text-red-500

color: rgb(239,68,68)

```
<p class="text-red-500">Red text</p>
```

leading-loose

line-height: 2

```
<p class="leading-loose">Airy  
lines</p>
```

Backgrounds, Borders, rounded-*, ring-*, divide-*



bg-blue-200

Light blue background

```
<div class="bg-blue-200 p-4">Blue BG</div>
```



rounded-full

border-radius: 9999px (circle)

```
<div class="rounded-full w-16 h-16">Circle</div>
```



border-4

4px solid border

```
<div class="border-4 border-gray-400">Box</div>
```



ring-2

Outline ring outside the border

```
<input class="ring-2 ring-purple-500">
```



border-dashed

Dashed border style

```
<div class="border border-dashed">Dashed</div>
```



divide-y

Borders between children

```
<div class="divide-y divide-gray-300">
```



rounded-lg

border-radius: 0.5rem

```
<div class="rounded-lg bg-pink-200">Rounded</div>
```



shadow-lg

Large box shadow

```
<div class="shadow-lg bg-white p-4">Card</div>
```

08

Effects & Visual Utilities

shadow-* opacity-* blur grayscale backdrop-blur

Effects — shadow, opacity, blur, grayscale

shadow-md

Medium box shadow

```
<div class="shadow-md p-4 bg-white">Card</div>
```

Lifts element visually off the page

shadow-lg

Large box shadow

```
<div class="shadow-lg p-6 bg-white">Big Card</div>
```

Stronger depth illusion

opacity-50

50% transparency

```

```

0=invisible 100=fully visible

blur

Blur the element

```
<div class="blur-sm">Blurry text</div>
```

blur-sm blur blur-md blur-lg

grayscale

Remove color

```

```

Use hover:grayscale-0 to restore color on hover

backdrop-blur

Blur element BEHIND

```
<div class="backdrop-blur-md bg-white/30">Glass</div>
```

Creates 'frosted glass' UI effect

Effects in Practice — Glassmorphism Card & Hover States

glass.html

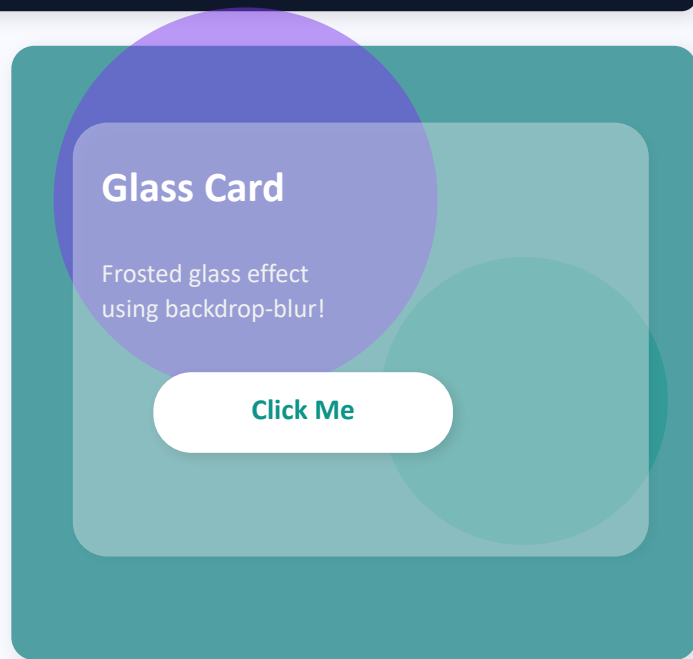
```
<!-- Glassmorphism card -->
<div class="min-h-screen bg-gradient-to-br from-teal-400
  to-purple-600 flex items-center justify-center">

  <div class="backdrop-blur-md bg-white/30 rounded-2xl
    p-8 shadow-2xl border border-white/40 w-80">
    <h2 class="text-2xl font-bold text-white mb-2">
      Glass Card
    </h2>
    <p class="text-white/80">
      This uses backdrop-blur to create a frosted glass
effect!
    </p>
    <button class="mt-4 px-6 py-2 bg-white text-teal-700
      rounded-full font-semibold hover:scale-105
      transition-transform shadow">

      Click Me
    </button>
  </div>
</div>

<!-- Image with grayscale + color on hover -->

```



← backdrop-blur-md + bg-white/30 creates this frosted glass!

Mini Projects — Card Component & Responsive Navbar

card.html

```
<!-- Card Component -->
<div class="max-w-sm rounded-2xl overflow-hidden
      shadow-lg bg-white">
  
  <div class="p-6">
    <h2 class="font-bold text-xl mb-2 text-
gray-800">
      Card Title
    </h2>
    <p class="text-gray-600 text-sm">
      A short description of the card content goes
here.
    </p>
    <button class="mt-4 w-full py-2 bg-teal-500
text-white
                    rounded-lg hover:bg-teal-600
font-semibold">
      Read More
    </button>
```

● ● ● Card Preview



Card Title

Short description of content...

Read More

navbar.html

```
<!-- Responsive Navbar -->
<nav class="bg-white shadow-md px-6 py-4">
  <div class="flex justify-between items-center
max-w-4xl mx-auto">

    <span class="text-xl font-bold text-
teal-600">
      MyBrand
    </span>

    <!-- Desktop links: hidden on mobile -->
    <ul class="hidden md:flex gap-6 text-
gray-600">
      <li class="hover:text-teal-500">Home</li>
      <li class="hover:text-teal-500">Projects</
li>
      <li class="hover:text-teal-500">Contact</
li>
    </ul>
```

● ● ● Navbar Preview

MyBrand

Home

Projects

Contact

Sign Up

desktop →

```
<!-- Hamburger (mobile only) -->
```

You've mastered

CSS Layout & Tailwind CSS

✓ Layout Basics (div, span, attributes)

✓ Block vs Inline elements

✓ Tailwind Utility-First approach

✓ CDN setup, Responsive utilities

✓ Dark Mode with dark: prefix

✓ Display, z-index, box-sizing

✓ Float, clear, overflow

✓ Flexbox (flex, justify, items, gap)

✓ Responsive Navbar with flex

✓ CSS Grid (cols, span, place)

✓ Spacing & Sizing (m, p, w, h)

✓ Typography & Text classes

✓ Backgrounds, Borders, rounded

✓ Ring, divide utilities

✓ shadow, opacity, blur effects

✓ Glassmorphism & hover states

Next up: JavaScript • Keep building! 🚀